Alp Berke Ardic

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SUMMARY

PhD candidate at UIC developing methods for decentralized LLM fine-tuning, distributed training, and model-parallel inference on edge devices. First author on all 6 papers, including an accepted NeurIPS 2026 workshop paper.

EDUCATION

University of Illinois Chicago, PhD in Electrical and Computer Engineering, GPA 4.0/4.0 — Aug 2022 – Present
Advised by Prof. Erdem Koyuncu. Teaching Assistant for Machine Learning, Deep Learning, and Pattern Recognition (100+ students).

University of Illinois Chicago, MS in Electrical and Computer Engineering, GPA 4.0/4.0 — Aug 2022 – Apr 2026

Bilkent University, BS in Electrical and Electronics Engineering, Ankara, Turkey — 2017 – 2022

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant, University of Illinois Chicago, Chicago, IL — Sep 2022 – Present

- Designed fully decentralized corrections for **participation bias** in gossip-based **LLM fine-tuning**, where nodes with rising loss train less often and their data becomes under-represented (accepted NeurIPS 2026 workshop paper).
- Built them in PyTorch with Hugging Face Transformers and PEFT, compared 19 methods, and tested the main ones with LoRA on pretrained LLMs. On the largest model the topology-aware correction never went unstable, unlike inverse participation weighting.
- Designed **Random Walking Snakes** for decentralized training, splitting a model across edge-device groups to train 40% faster than random-walk learning, then extended it to transformer training on non-IID data (IEEE LANMAN 2023, IEEE TMC under review).
- Created coordinator-free **model-parallel inference** that assigns DNN layer blocks to edge devices by decentralized graph coloring, cutting assignment conflicts by over 90% and inference latency by 10% (IEEE MILCOM 2025, IEEE TPDS under review).

Research Intern, TUBITAK SAGE (national defense R&D institute), Ankara, Turkey — Sep 2021 – May 2022

- Raised real-time YOLO object detection accuracy by 18% under motion blur using custom augmentation and hyperparameter tuning.
- Sped up the inference preprocessing pipeline by 25% for real-time, high-speed imaging.

Robotics Researcher, Bilkent Miniature Robotics Lab | inspection UAVs, reverse-engineered commercial drones — Jan 2021 – Nov 2021

Embedded and Camera Systems Intern, ASELSAN | camera interface built from scratch in C on a microcontroller — Jun 2021 – Aug 2021

AI and Computer Vision Intern, TWIN Science & Robotics | real-time CNN vision on Raspberry Pi — Jul 2019 – Sep 2019

SELECTED PROJECTS

On-Device CNN Compression for ESP32 | ECE 594 team project, UIC — Oct 2025 – Dec 2025

- Led the model and compression design (structured pruning, adaptive pooling, INT8 quantization) and wrote the Python training code, shrinking a CIFAR-10 CNN 42× to run on an ESP32 microcontroller while keeping over 95% of its accuracy.

Reinforcement Learning MAC Protocol in ns-3 | ECE 533 solo project, UIC — Oct 2025 – Dec 2025

- Implemented RL-MAC in ns-3 (C++) and a Python Q-learning agent that earned up to 2× the reward of random and fixed-slot baselines.

PUBLICATIONS

- **A. B. Ardic**, E. Koyuncu, "Who Gets Heard by Gossip? Correcting Participation Bias in Decentralized Language-Model Fine-Tuning," *NeurIPS 2026 Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM)*, accepted.
- **A. B. Ardic**, E. Koyuncu, "When Does Spatial Cooperation Help? Contamination Regimes in Decentralized Online Spectrum Anomaly Learning," *IEEE MILCOM 2026 Workshop*, accepted.
- **A. B. Ardic**, M. Dwyer, E. Koyuncu, "Decentralized Model-Distributed Inference," *IEEE MILCOM 2025*.
- **A. B. Ardic**, H. Seferoglu, S. El Rouayheb, E. Koyuncu, "Random Walking Snakes for Decentralized Learning at Edge Networks," *IEEE LANMAN 2023*.
- **A. B. Ardic**, M. Dwyer, E. Koyuncu, "Cost-Aware Asynchronous Persistent Graph Coloring for Decentralized Model-Distributed Inference," *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, under review.
- **A. B. Ardic**, H. Seferoglu, S. El Rouayheb, E. Koyuncu, "Random Walking Snakes for Decentralized Learning of Heterogeneous Data at Edge Networks," *IEEE Transactions on Mobile Computing (TMC)*, under review.

TECHNICAL SKILLS

Machine Learning | large language models (LLMs), parameter-efficient fine-tuning (LoRA), distributed training algorithms, decentralized optimization, federated learning, model-parallel inference, model compression, computer vision, reinforcement learning

Frameworks | PyTorch, Hugging Face Transformers, PEFT, TensorFlow, OpenCV

Languages and Tools | Python, C/C++, MATLAB, Linux, ns-3, Raspberry Pi, Arduino